Molecular Characterization Laboratory
Frederick National Laboratory for Cancer Research
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Patient Name:		Patient DOB:	
Date Collected:		Referring Physician:	
Biopsy Site:	**Bone marrow**	Primary Diagnosis:	**Acute myeloid leukemia**
Sample ID:		MoCha ID:	
Telephone:		Fax:	
Tumor Content (%):	**2% blast**		

Preanalytics to include tumor enrichment as required is performed by Roswell Park Comprehensive Cancer Center; Buffalo; NY

14203

Results:f
No mutations are identified in this patient's specimen.

Comments:
N/A

Report Signed By

Name	Role	Date

Assay Information

Methodology, Scope, and Application:The NCI Myeloid Assay v2 (NMAv2) is a next-generation sequencing (NGS) assay which identifies pre-defined and novel genomic variants covering 45 DNA genes and 35 fusion drivers that are categorized into 3 mutation types: single nucleotide variants (SNVs), insertions/deletions (Indels) including FLT3 internal tandem duplications (ITDs), and gene fusions. The assay utilizes Thermo Fisher Scientific's Oncomine® Myeloid Assay GX v2, a NGS assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from blood and bone marrow mononuclear cells for sequencing on the Ion Torrent Genexus Integrated Sequencer and analyzed by the Ion Torrent Genexus pipeline version 6.6.2.1. The NMAv2 can reliably identify the presence or absence of 1661 hotspot variants and 779 reported fusions compared to the Human Reference Genome hg19, including the genes listed below. The NMAv2 is a laboratory developed test designed to detect gene mutations for major myeloid disorders.

Analytical Sensitivity and Specificity:The NMAv2 has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 98.97% sensitivity, 100% specificity and an overall reproducibility with a mean positive percent agreement of 99% and a mean negative percent agreement of 100% during validation of the assay.

Hereditary and Germline Mutations:This assay examines cancer cells only and does not examine normal (non-cancer) cells. Mutations detected by the assay may be present only in the cancer cells, or in every cell of the body (including non-cancer cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or more features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's), it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-cancer) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content:Thermo Fisher Scientific's Ion Torrent Oncomine Knowledgebase Reporter (OKR) software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OKR. The data presented here is from a curated knowledgebase of publicly available information but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:
ABL1, ANKRD26, BRAF, CBL, CSF3R, DDX41, DNMT3A, FLT3, GATA2, HRAS, IDH1, IDH2, JAK2, KIT, KRAS, MPL, MYD88, NPM1, NRAS, PPM1D, PTPN11, SETBP1, SF3B1, SMC1A, SMC3, SRSF2, U2AF1, WT1

Genes assayed with full exon coverage:
ASXL1, BCOR, CALR, CEBPA, ETV6, EZH2, IKZF1, NF1, PHF6, PRPF8, RB1, RUNX1, SH2B3, STAG2, TET2, TP53, ZRSR2

Genes assayed for detection of fusions:
ABL1, ABL2, BCL2, BRAF, CCND1, CREBBP, EGFR, ETV6, FGFR1, FGFR2, FUS, HMGA2, JAK2, KAT6A, KAT6B, KMT2A, KMT2A-PTD (Partial Tandem Duplication), MECOM, MET, MLLT10, MRTFA (MKL1), MYBL1, MYH11, NTRK2, NTRK3, NUP214, NUP98, PAX5, PDGFRA, PDGFRB, RARA, RUNX1, TCF3, TFE3, ZNF384

Limitaions of the assay

1. Mutations in homopolymer regions of >5 nucleotides may not be accurately assessed by this assay.
2. FLT3-ITDs greater than 120bps may not be detected by this assay. Sensitivity for larger FLT3-ITD's may be reduced and variant allele frequencies may be reported lower than expected.
3. Following genetic alterations will not be reported by this assay: RUNX1 p.Ser445AlafsTer149, ASXL1 p.G646Wfs*12, PRPF8 p.Arg156GlufsTer28

DISCLAIMER

This assay reports SNV, indel and FLT3-ITD mutations ≥ 5% Variant Allele Frequency (VAF). This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.